

Shigella

**Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health**

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Highlights

- There is strong evidence linking transmission of *Shigella* with unsafe drinking-water, inadequate sanitation, and poor hygiene. *Shigella* are major waterborne pathogens. Multiple drinking-water outbreaks caused by *Shigella* have been documented.
- Humans are the main host and reservoir for *Shigella*, but higher primates may also be potential sources. *Shigella* are generally considered to be part of the same pathotype as enteroinvasive *E. coli* but are more virulent.
- Transmission is primarily by the faecal–oral route through person-to-person contact and ingestion of contaminated food and water.
- *Shigella* may cause mild to severe intestinal infections, including dysentery. Complications include toxic megacolon, intestinal perforations and haemolytic uraemic syndrome. Antibiotic treatment is an important response to dysentery. However, this is becoming compromised by high levels of resistance to the first line antibiotics.
- Shigellosis disproportionately impacts low-income countries. Elderly, malnourished, and immunocompromised individuals and children are particularly susceptible to severe disease and death.
- Infectivity of *Shigella* varies from highly infective for *S. dysenteriae* with an estimated ID₅₀ (50% infective dose) of 36 organisms to moderately infective for *S. flexneri* with an estimated ID₅₀ of 148 organisms.
- Standard culture-based methods to detect and differentiate *Shigella* in environmental samples are available. Molecular methods are also available.
- In a small number of studies, *Shigella* have been detected in groundwater and drinking-water contaminated by sewage. Based on limited data, it is likely that *Shigella* are moderately persistent in drinking-water sources.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from *Shigella*.
- Reduction of *Shigella* by wastewater and drinking-water treatment varies depending on the process but is similar to the reduction of faecal indicator bacteria such as *E. coli*. Disinfection processes including chlorination, are effective for inactivating *Shigella*.
- *Escherichia coli* (or alternatively, thermotolerant coliforms) is a suitable indicator for the possible presence or absence of *Shigella* in drinking-water.

1 General description

Shigella are members of the Enterobacteriaceae family. They are Gram-negative, non-spore-forming, non-motile, rod-like bacteria, 0.5 µm by 1 to 3 µm in size.

There are four separate species of *Shigella*: *S. dysenteriae*, *S. flexneri*, *S. boydii* and *S. sonnei*, which are further divided into a number of serotypes. The antigenic classification of the genus *Shigella* is based essentially on their somatic O antigens, many of which are shared with other enteric bacilli. Although *Shigella* is classified as a separate genus, it is genetically very similar to pathogenic *Escherichia coli* and it is generally considered that they belong to the same pathotype as enteroinvasive *E. coli* (EIEC) but is more virulent (Niyogi, 2005; Croxen et al., 2013; Garcia-Aljaro et al., 2017). EIEC is the subject of a separate background document.

2 Human health effects

Shigella are highly invasive intracellular pathogens that can cause serious intestinal infections, including bacillary dysentery (shigellosis) (Croxen et al., 2013; Garcia-Aljaro et al., 2017). The severity and case fatality vary by species and serotype, as well as host characteristics (Heymann, 2015). Symptoms of *Shigella* infections are similar to those produced by enteroinvasive *E. coli*. They share many of the same virulence factors and biochemical characteristics, but *Shigella* exhibit greater expression of virulence resulting in more severe disease (Croxen et al., 2013; Garcia-Aljaro et al., 2017). One important difference is the production of Shiga toxin, mostly by *S. dysenteriae*.

Abdominal cramps, fever, watery diarrhoea and dehydration occur early in the infection. However, severe and life-threatening complications can occur, particularly in low-income countries where access to adequate treatment is limited. Infections of *S. dysenteriae* may proceed to an ulceration process, with bloody diarrhoea and high concentrations of neutrophils in the stool. The production of Shiga toxin by the pathogen plays an important role in this outcome (Croxen et al., 2013). In the elderly and in children under 5 years of age, shigellosis may result in severe dehydration and death.

Shiga toxin producing *S. dysenteriae* type 1 is most often associated with epidemics of severe disease and associated complications, including toxic megacolon, intestinal perforations and haemolytic uraemic syndrome (Croxen et al., 2013). Case fatality rates as high as 15% among hospitalized cases have been reported (WHO, 2005). Children hospitalized with shigellosis frequently experience seizures. Long-term complications in children infected with *S. dysenteriae* in low- and middle-income countries include persistent diarrhoea and malnutrition (Kotloff et al., 2018). *Shigella flexneri* has been associated with infectious arthritis (Martin et al., 2012). Illness due to *S. sonnei* is usually relatively mild and self-limiting.

The choice of antibiotics for treating dysentery is becoming limited due to the global emergence of antimicrobial resistance. A high incidence of resistance of *Shigella* to all first line antibiotics, including sulphonamides, ampicillin, tetracycline, chloramphenicol, sulfamethoxazole/trimethoprim and nalidixic acid, has been reported (Croxen et al., 2013). Fluoroquinolones, cephalosporin derivatives and azithromycin are recommended (WHO, 2005; Croxen et al., 2013).

3 Epidemiology

Shigella is one of the most common aetiological agents of moderate to severe diarrhoea and death, including among children under 5 years of age in low- and middle-income countries (Platts-Mills et al., 2015; GBD, 2020).

Shigellosis can be endemic in both temperate and tropical climates and epidemics or sporadic infection can also occur. In the past 40 years, four *S. dysenteriae* type 1 pandemics have occurred resulting in high rates of fatalities for all age groups in Central America, South Asia, Central Africa and East Africa (Garcia-Aljaro et al., 2017).

The geographical distribution of the four *Shigella* species is different; *S. dysenteriae* is mainly found in densely populated areas of South America, Africa and Asia, *S. flexneri* is the most frequently isolated species and predominates in low-income countries with endemic shigellosis, *S. boydii* occurs sporadically except for in India and *S. sonnei* is mostly reported in high-income countries, such as Europe and North America (Croxen et al., 2013; Garcia-Aljaro et al., 2017). Seasonal fluctuations in the number of shigellosis outbreaks have been demonstrated in many countries (Engels et al., 1995; McQuade et al., 2020).

4 Transmission

Shigella are transmitted primarily by the faecal–oral route through person-to-person contact, and ingestion of contaminated food or water. Flies have also been identified as a transmission vector from contaminated faecal waste (Khalil et al., 1994). Humans and other higher primates appear to be the only natural hosts for *Shigella* (Croxen et al., 2013; Garcia-Aljaro et al., 2017). The risk of transmission is greatest in low-income settings where sanitation and hygiene conditions are generally poor. Sporadic cases and outbreaks of shigellosis often occur in crowded communities or institutions with inadequate sanitation and poor hygiene, for example, in day-care centres, prisons, residential health-care settings or military field operations (Kotloff et al., 1999; Garcia-Aljaro et al., 2017). Secondary attack rates in households can be as high as 40% (Heymann, 2015). Outbreaks of shigellosis have also been associated with recreational bathing waters (Garcia-Aljaro et al., 2017).

The relative infectivity varies among the species. *Shigella dysenteriae* is highly infective¹ with an estimated ID₅₀ (50% infectious dose i.e. the number of organisms required to infect 50% of exposed subjects) of 36 organisms. *Shigella flexneri* is moderately infective with an estimated ID₅₀ of 148 organisms (Powell, 2000). The incubation period ranges between 1 and 3 days but may be up to 1 week depending on the species and the host (Heymann, 2015). The onset of symptoms is usually sudden and typically lasts for 4 days. *Shigella* is communicable during acute infection and as long as the infectious agent is actively shed in stool. Infected humans are estimated to excrete 10⁶ to 10⁸ cells/gram of stool (Gaurav et al., 2013). The pathogen is shed usually for up to 4 weeks during the acute illness and in the post-infection asymptomatic phases,

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: High means infective doses can be 1–10² organisms or particles, Moderate 10²–10⁴ and Low > 10⁴.

but in rare cases shedding can continue for multiple months (Heymann, 2015; Garcia-Aljaro et al., 2017). Several studies indicate that repeated exposures to *Shigella* can provide protective immunity (Niyogi, 2005).

5 Detection methods

Standard culture-based methods to detect and differentiate *Shigella* in environmental samples are available (e.g. ISO 21567, 2004) but the methods are complex and involve multiple steps. In addition, commercial biochemical tests to type isolates exist. Molecular methods are available also, are less onerous than culture-based methods and have a comparable sensitivity (Dutta et al., 2001; Islam et al., 1998; Hounig et al., 1997). However, distinguishing between viable and non-viable cells is a shortcoming of molecular methods.

6 Source and occurrence

Data on the occurrence of *Shigella* in environmental media are scarce due to the difficulties with the detection and identification of these pathogens in environmental samples (Garcia-Aljaro et al., 2017). Similarly, data on survival of *Shigella* in water are scarce.

6.1 Faecal pollution sources

Shigella have been identified in raw sewage but very few studies have quantified their concentrations (Garcia-Aljaro et al., 2017). Survival times for *Shigella* in faeces have been reported to be up to 12 days at 25 °C (Islam et al., 1993). Flies can carry *Shigella* for up to 20 to 24 days (Khalil et al., 1994).

There are few studies of *Shigella* prevalence in treated wastewater (Garcia-Aljaro et al., 2017). *Shigella* were detected in up to 60% of wastewater effluent samples in South Africa, in concentrations of 0.4 log₁₀ colony-forming units (CFU)/mL to 6.3 log₁₀ CFU/mL (Teklehaimanot et al., 2015). In laboratory experiments using domestic wastewater treatment plant effluent at 4 °C and 25 °C, inocula of 10⁹ cells/mL were reduced by 2 to 3 log₁₀ in 24 hours, followed by a more gradual reduction leading to a total loss of 5 log₁₀ over 30 days (Ellafi et al., 2010).

6.2 Surface water

Shigella have been detected in lakes, rivers, streams and recreational waters in low- and high-income settings (Garcia-Aljaro et al., 2017). Data are very limited on concentrations in surface waters; one study detected 730 CFU/L in one of 30 samples (Hsu et al., 2008). As noted above, data on the persistence of *Shigella* in waters are scarce. In a laboratory microcosm study conducted at ambient temperatures, 10⁶ cells/mL of *S. flexneri* were not culturable after 7, 8 and 10 days in lake water, pond water and tap water, respectively (Sultana et al., 2005). In addition, Mitscherlich & Marth (1984; as cited by Sorvillo et al., 1988) reported 3 days survival in a recreational surface water. Based on these limited studies, persistence of *Shigella* is likely

moderate². Nevertheless, there is some indication that *Shigella* can survive for longer periods in fresh water (Mitscherlich & Marth 1984; as cited by Sorvillo et al., 1988).

6.3 Groundwater and drinking-water

Data on the presence of *Shigella* in groundwater and drinking-water are very limited. Using molecular methods Mpenyana-Monyatsi & Momba (2012) determined that one of 100 boreholes in South Africa contained *S. dysenteriae*. In addition, *S. sonnei* was detected by a culture-based method in a drinking-water sample coinciding with an outbreak in Greece linked to contaminated groundwater (Alamanos et al., 2000).

6.4 Soil and food

Shigella have been detected in soil samples, fruits, vegetables and herbs, and in fish and shellfish from seawater and freshwater. *Shigella* are acid resistant and salt-tolerant and may survive on foods maintained in these conditions (Garcia-Aljaro et al., 2017). Survival times for *Shigella* have been reported up to 10 days in citric juices and carbonated soft drinks and several days on contaminated vegetables (Islam et al., 1993; Downes and Ito, 2001).

6.5 Surfaces

Survival times for *Shigella* have been reported as between 2 days to 5 months on dry surfaces, over 3 hours on fingers, 2 to 28 days on metal utensils at 15 °C, or up to 13 days at 37 °C (Kramer et al., 2006).

7 Significance of sanitation

Safe sanitation systems are an important barrier against the spread of *Shigella* from human faeces. Sewage has been identified as a source of several drinking-water outbreaks (Green et al., 1968; PAHO, 2004; Alamanos et al., 2000), and an epidemiological study in Argentina demonstrated that poor sanitation and drinking-water quality were related to an increased risk of shigellosis (Rolfo et al., 2012). Raw vegetables irrigated with contaminated water have also been implicated in *Shigella* outbreaks, suggesting that adequate collection, containment and treatment of faecal waste as well as protection of water sources used for irrigation, may successfully disrupt this transmission pathway (Bennett et al., 2018).

8 Significance in drinking-water

Shigella are major waterborne pathogens. There is strong evidence linking transmission of *Shigella* with unsafe drinking-water. Multiple drinking-water outbreaks have been documented. For example, a large waterborne outbreak of *S. sonnei* occurred in a town in Scotland in 1966 and affected approximately 2000 residents (Green et al., 1968). An outbreak affecting 288 people in a village in Greece in 2000 was attributed to *S. sonnei* in a contaminated groundwater

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

supply (Alamanos et al., 2000) while a large *S. flexneri* waterborne outbreak occurred in 2004 in a small city of Argentina and affected more than 3100 inhabitants (PAHO, 2004).

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from *Shigella*.

The effectiveness of sanitation and drinking-water treatment systems depend on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on size exclusion (e.g. by filtration through media or soil in septic tank drainage areas) or settling in primary wastewater treatment processes will be limited for removing bacteria, including *Shigella*. Oxidizing chemical disinfectants, such as chlorine, and UV light, are effective at inactivating non-spore forming enteric bacteria such as *Shigella*.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Data on removal or inactivation of *Shigella* by wastewater treatment processes are limited (Garcia-Aljaro et al., 2017). However, given that *Shigella* belongs to the same pathotype as EIEC, the reduction of *Shigella* by wastewater treatment processes is likely to be similar to the reduction of faecal indicator bacteria such as *E. coli*.

On-site sanitation systems should reduce *Shigella* concentrations, but the extent will depend on the type of system and environmental factors. A 1 log₁₀ removal of *Shigella* can be expected in well-constructed septic tanks. Thermophilic composting and storage can reduce *Shigella* concentrations; composting waste at 55 °C to 60 °C for one month, followed by a 2- to 4-month maturation period, ensures sufficient bacterial reductions (Strauss & Blumenthal, 1990).

In centralized sewage treatment plants, bacteria such as *Shigella* are not very sensitive to removal by size exclusion or settling processes, and it is expected that clarifiers or sedimentation tanks will generally achieve less than 1 log₁₀ reduction in concentrations (Oakley, 2018). Based on faecal indicator bacteria the majority of bacterial removal occurs during secondary treatment; trickling filters, activated sludge treatment and membrane bioreactors should produce greater than 2 log₁₀ reductions (Garcia-Aljaro et al., 2017).

9.2 Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023), control measures to manage the potential risk from *Shigella* should include protection of source waters (surface or groundwater) from contamination by human faecal waste (including extra vigilance during heavy rain), application of adequate water treatment processes and protection of drinking-water quality during storage

and distribution. *Shigella* have a low resistance³ to disinfectants such as chlorine and UV light. Ct (product of disinfectant concentration and contact time) values of 0.4 to 0.5 mg.min/L for chlorine at 25 °C and pH 7 achieved a 2 log₁₀ inactivation (King et al., 1988). A UV light dose of 5 to 13 mJ/cm² also achieved a 2 log₁₀ inactivation of *Shigella* (Hijnen et al., 2006).

Escherichia coli (or alternatively, thermotolerant coliforms) is a suitable indicator for the possible presence or absence of *Shigella* in drinking-water.

10 ICD-11 classification

XN7HG	<i>Shigella</i> spp.
XN7Y2	<i>Shigella flexneri</i>
XN8RN	<i>Shigella boydii</i>
XN285	<i>Shigella dysenteriae</i>
XN9M9	<i>Shigella sonnei</i>
1A02	Intestinal infections due to <i>Shigella</i>
MG50.A	Antibiotic resistant <i>Shigella</i>

³ Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Resistance is based on 99% inactivation at 20 °C where, generally, Low represents a Ct99 of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8. It should be noted that within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Organisms that survive and grow in biofilms will be protected from chlorination.

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Shigella

Background document for the WHO Guidelines for drinking-water quality
and the WHO Guidelines on sanitation and health

[WHO \(2023\). Water safety plan manual: Step-by-step risk management for drinking-water suppliers, second edition. Geneva: World Health Organization.](#)

Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.